

Matching for Causal Inference Without Balance Checking

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joint work with
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(talk at the University of Notre Dame, 1/29/09)

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- Removing heterogeneous data will often reduce variance too
- (Medical experiments are the reverse: small- n with random treatment assignment; don't match unless something goes wrong)

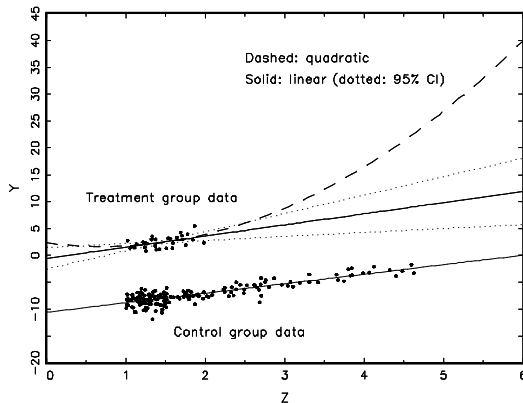
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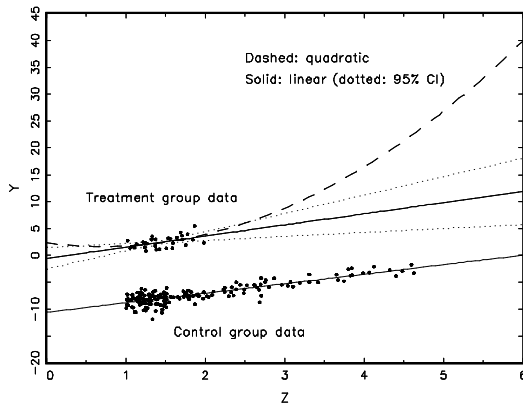
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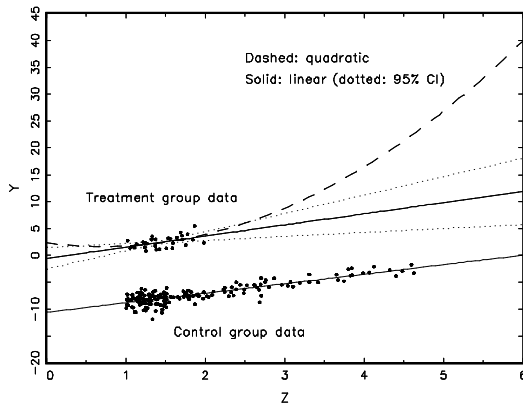
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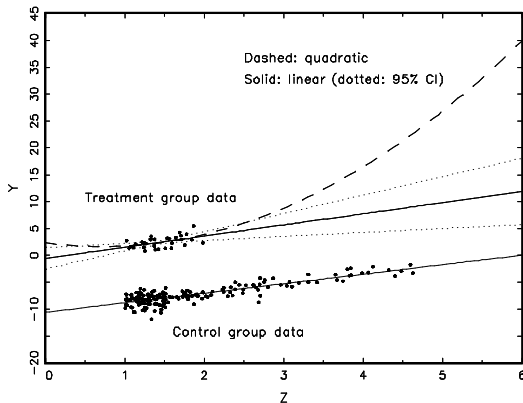


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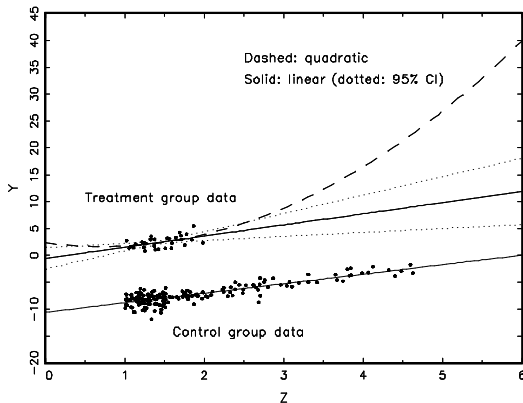


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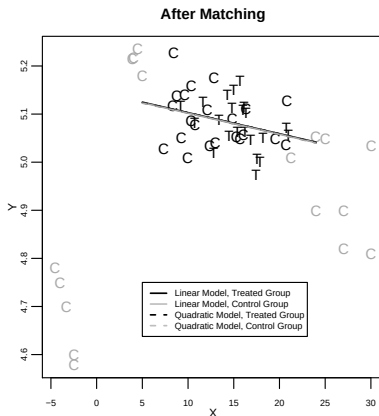
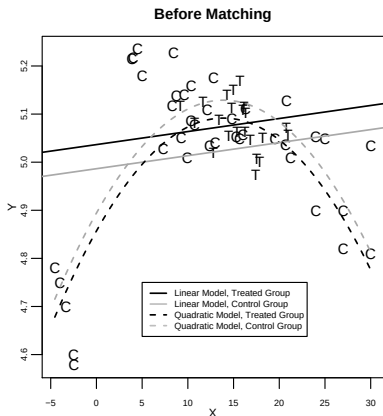
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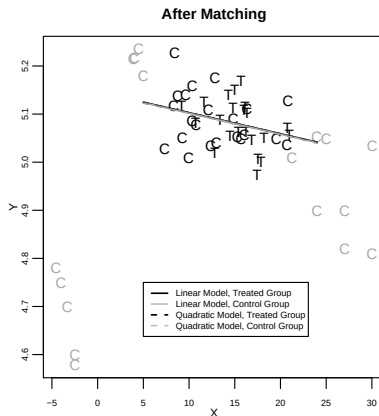
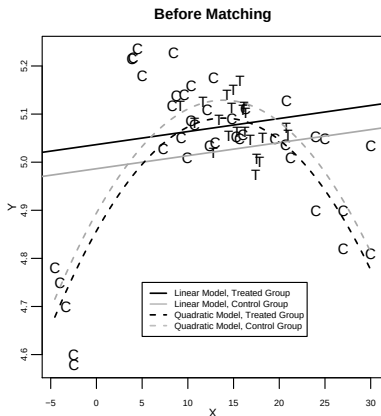
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Matching reduces model dependence, bias, and variance

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$$\text{SATT} = \frac{1}{n_T} \sum_{i \in \{T_i=1\}} \text{TE}_i$$

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(Is balance even improved?)

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If ϵ is reduced, $\gamma(\epsilon)$ decreases & $\gamma(\epsilon)$ is unchanged

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- What if ϵ is set ...

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- We Prove: setting ϵ bounds the treated-control group difference, within strata and globally, for: means, variances, skewness, covariances, comoments, coskewness, co-kurtosis, quantiles, and full multivariate histogram.
 - ⇒ Setting ϵ controls all multivariate treatment-control differences, interactions, and nonlinearities, up to the chosen level (matched n is determined ex post)
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- Simple to teach: coarsen, then exact match

Imbalance Measures

Variable-by-Variable Difference in Global Means

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Local Imbalance by Variable (given strata fixed by matching method)

$$I_2^{(j)} = \frac{1}{S} \sum_{s=1}^S \left| \bar{X}_{m_T^s}^{(j)} - \bar{X}_{m_C^s}^{(j)} \right|, \quad j = 1, \dots, k$$

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⇒ **CEM dominates EPBR-methods in EPBR Data**

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⇒ CEM works well in non-EPBR data too

CEM Extensions I

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For papers, software (for R and Stata), tutorials, etc.

<http://GKing.Harvard.edu/cem>